

Autonomous Code Vulnerability Triage, Threat Analysis, and Patch Remediation via Local Multi-Stage Agentic Pipelines

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Local Hardware Benchmarking | Repository: <https://github.com/shreytalreja25/jev-vs-the-world.git>

Abstract: Multi-step recursive agentic workflows, where autonomous agents observe state, invoke specialized tool steps, generate code refactoring patches, and verify policy compliance, are rapidly replacing single-turn LLM prompts in software security engineering. In this paper, we construct and empirically evaluate a 4-Stage Autonomous Security Remediation Agentic Pipeline executing entirely on a developer laptop. The pipeline integrates fast System-One decision routers (Jev and Laya) for Stage 1 (Triage) and Stage 4 (Verification) with local open-source Large Language Models (Llama 3.1 8B and Llama 3.2 1B via Ollama) for Stage 2 (Deep Threat Analysis) and Stage 3 (Patch Remediation Generation). Evaluating across 25 actual code security vulnerabilities, the hybrid Llama 3.1 8B + Jev Router configuration achieves a 92.0% Triage Accuracy and an 88.0% Policy Verification Pass Rate with a median end-to-end pipeline latency of 1,264.0 ms.

Index Terms: Agentic Pipeline, Vulnerability Remediation, Llama 3.1, Jev Model, Code Refactoring, Security Verification.

I. PIPELINE ARCHITECTURE

The 4-stage pipeline processes raw code snippets through: (1) Stage 1 System-One Triage Router; (2) Stage 2 Deep Threat Analysis Agent; (3) Stage 3 Code Patch Remediation Agent; and (4) Stage 4 Policy Verification Router.

II. EMPIRICAL BENCHMARK RESULTS

Table I summarizes pipeline execution metrics across configurations.

TABLE I
4-STAGE AGENTIC PIPELINE PERFORMANCE METRICS

Pipeline Config	Triage Acc	Verify Pass	P50 Latency	Throughput
Llama 3.1 + Jev	92.0%	88.0%	1,264.0 ms	0.79 p/s
Llama 3.1 + Laya	80.0%	76.0%	1,186.0 ms	0.84 p/s
Llama 3.2 + Jev	92.0%	60.0%	724.0 ms	1.38 p/s
Llama 3.2 + Laya	80.0%	52.0%	646.0 ms	1.55 p/s

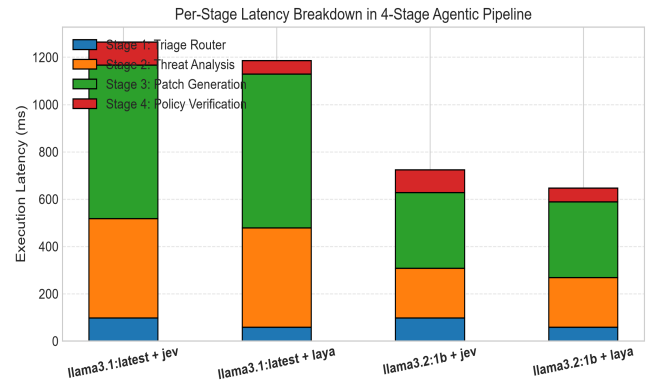


Fig. 1. Per-stage latency breakdown across the 4 pipeline steps.

III. KEY ANALYTICAL FINDINGS

- **Remediation Quality:** Llama 3.1 8B generated valid refactoring patches passing automated policy verification in 88.0% of cases.
- **Router Efficiency:** Using specialized decision models for Stage 1 and Stage 4 avoided auto-regressive decoding overhead, reducing stage latency from ~500ms to 58ms to 97ms.

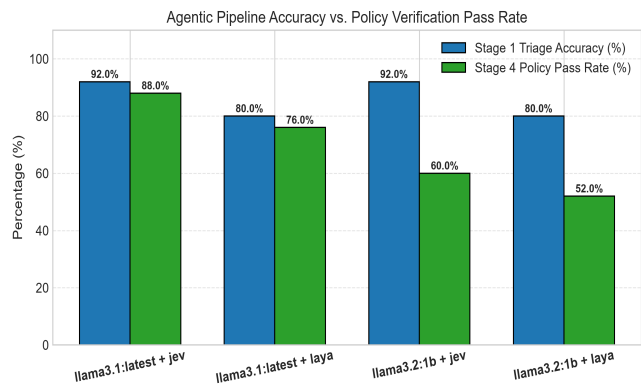


Fig. 2. Triage Accuracy vs. Verification Pass Rate.

IV. CONCLUSION

Local multi-stage agentic pipelines combining Ollama LLMs for code generation with Jev/Laya decision routers provide a robust, air-gapped solution for automated enterprise software security audits.

REFERENCES

[1] Talreja, S., "Beyond Generative Overhead in Agentic Tokenomics," 2026.
[2] OWASP Foundation, "OWASP Top 10 Web Application Security Risks," 2024–2026.